

Network+ N10-009 Memorization Sheet

Print this and keep it beside you while you study. Every number, order, and pairing on this sheet is fair game on the exam. Nothing here needs understanding to score; it needs recall.

01 The exam itself

FACT	VALUE
Exam code	N10-009 (Network+)
Questions	Up to 90, multiple choice plus performance-based
Time	90 minutes
Passing score	720 on a scale of 100 to 900 (about 72%)
1.0 Networking Concepts	23%
2.0 Network Implementation	20%
3.0 Network Operations	19%
4.0 Network Security	14%
5.0 Network Troubleshooting	24%

Read for the qualifier: MOST likely, BEST, FIRST, NEXT. Eliminate two options, then choose. Never leave a question blank.

02 Ports and protocols

SERVICE	PORT	TRANSPORT	NOTES AND SECURE FORM
FTP	20 data, 21 control	TCP	Plaintext. Secure: SFTP on 22, or FTPS on 989/990
SSH, SFTP, SCP	22	TCP	Encrypted shell and file copy. Replaces Telnet and FTP
Telnet	23	TCP	Plaintext. Replace with SSH
SMTP	25	TCP	Server to server mail. Secure submission: 587
TACACS+	49	TCP	Device administration AAA, encrypts the whole payload
DNS	53	UDP queries, TCP zone transfers	Secure: DoT 853, DoH 443
DHCP	67 server, 68 client	UDP	DORA. DHCPv6 uses 546 client, 547 server
TFTP	69	UDP	No authentication. Firmware and config transfers
HTTP	80	TCP	Secure: HTTPS 443
Kerberos	88	TCP or UDP	Windows domain authentication. Breaks if clocks differ by more than 5 minutes
POP3	110	TCP	Secure: POP3S 995
NTP	123	UDP	Time sync. Secured with NTS
IMAP	143	TCP	Secure: IMAPS 993
SNMP	161 agent (polls), 162 manager (traps)	UDP	v1 and v2c use a community string; v3 adds authentication and encryption

SERVICE	PORT	TRANSPORT	NOTES AND SECURE FORM
LDAP	389	TCP	Secure: LDAPS 636
HTTPS	443	TCP	TLS web traffic; also carries DoH
SMB	445	TCP	Windows file and printer sharing
IKE (IPsec)	500, NAT-T 4500	UDP	Key exchange. ESP is IP protocol 50, AH is 51
Syslog	514	UDP	Central logging. Severity 0 to 7
SMTPS, submission	587	TCP	Mail client to server with TLS
LDAPS	636	TCP	LDAP over TLS
DNS over TLS	853	TCP	Encrypted resolver queries
IMAPS	993	TCP	IMAP over TLS
POP3S	995	TCP	POP3 over TLS
SQL Server	1433	TCP	Microsoft SQL
RADIUS	1812 auth, 1813 accounting	UDP	Network access AAA (Wi-Fi, VPN, 802.1X). Legacy 1645/1646
MySQL	3306	TCP	MySQL and MariaDB
RDP	3389	TCP	Remote Desktop
SIP	5060 plain, 5061 TLS	TCP or UDP	Sets up voice and video calls

Encrypted pairs

- HTTP 80 to HTTPS 443
- Telnet 23 to SSH 22
- FTP 21 to SFTP 22
- LDAP 389 to LDAPS 636
- IMAP 143 to IMAPS 993
- POP3 110 to POP3S 995
- SMTP 25 to SMTPS 587
- DNS 53 to DoT 853 or DoH 443
- SNMP v1/v2c to SNMPv3 (same ports)
- SIP 5060 to SIP over TLS 5061

Port ranges and IP protocol numbers

- 0 to 1023: well-known
- 1024 to 49151: registered
- 49152 to 65535: dynamic (client) ports
- ICMP 1, TCP 6, UDP 17
- GRE 47, ESP 50, AH 51
- OSPF 89, EIGRP 88 (ride directly on IP)

03 OSI and TCP/IP models

LAYER	PDU	KEY IDEA	DEVICES AND PROTOCOLS
7 Application	Data	Protocols users touch	HTTP, HTTPS, SMTP, DNS, DHCP, FTP, SSH, SNMP, Telnet; proxies, web application firewalls
6 Presentation	Data	Formatting, encryption, compression	TLS, JPEG, MPEG, ASCII, Unicode
5 Session	Data	Start, maintain, end dialogs	NetBIOS, RPC, session control
4 Transport	Segment (TCP), datagram (UDP)	Port numbers; reliable vs fast	TCP, UDP; port-filtering firewalls, load balancers
3 Network	Packet	IP addresses, TTL, routing	Routers, Layer 3 switches; IP, ICMP, IPsec, OSPF, RIP

LAYER	PDU	KEY IDEA	DEVICES AND PROTOCOLS
2 Data Link	Frame	MAC addresses, MAC and LLC sublayers, frame check sequence	Switches, NICs, access points, bridges; Ethernet, 802.11, ARP, STP, 802.1Q
1 Physical	Bit	Signals, cables, connectors, radio	Hubs, repeaters, transceivers, media converters, cables

- Bottom up: Please Do Not Throw Sausage Pizza Away
- Top down: All People Seem To Need Data Processing
- TCP/IP model: Application (OSI 5 to 7), Transport (4), Internet (3), Network Access or Link (1 and 2)
- Encapsulation: data, segment, packet, frame, bits. Capture order: Ethernet header (MACs), IP header (addresses, TTL), TCP or UDP header (ports), payload
- MAC address: 48 bits, 12 hex digits, first 24 bits are the OUI. Rewritten at every router hop; the IP address stays the same end to end
- MTU 1500 bytes standard; jumbo frames about 9000
- TCP handshake: SYN, SYN-ACK, ACK. Close: FIN and ACK. Abort: RST
- Traffic types: unicast one to one; broadcast one to all on the segment (routers stop it; IPv6 has none); multicast one to a group (224.0.0.0 to 239.255.255.255); anycast one address on many servers, nearest answers

04 IPv4 addressing and subnetting

CLASS	FIRST OCTET	DEFAULT MASK	USE
A	1 to 126	/8, 255.0.0.0	Huge networks
B	128 to 191	/16, 255.255.0.0	Medium networks
C	192 to 223	/24, 255.255.255.0	Small networks
D	224 to 239	none	Multicast (OSPF uses 224.0.0.5 and 224.0.0.6)
E	240 to 255	none	Experimental
127	127.0.0.0/8	loopback	127.0.0.1 is this host

RANGE	MEANING
10.0.0.0/8 (10.0.0.0 to 10.255.255.255)	Private, RFC 1918
172.16.0.0/12 (172.16.0.0 to 172.31.255.255)	Private, RFC 1918. 172.32.x.x is public
192.168.0.0/16 (192.168.0.0 to 192.168.255.255)	Private, RFC 1918
169.254.0.0/16	APIPA, link-local. Means DHCP failed
255.255.255.255	Limited broadcast (DHCP Discover)
0.0.0.0/0	Default route, route of last resort
100.64.0.0/10	Carrier-grade NAT shared space

PREFIX	MASK	BLOCK SIZE	USABLE HOSTS
/8	255.0.0.0	1 in octet 1	16,777,214
/16	255.255.0.0	1 in octet 2	65,534
/17	255.255.128.0	128 in octet 3	32,766
/18	255.255.192.0	64 in octet 3	16,382
/19	255.255.224.0	32 in octet 3	8,190
/20	255.255.240.0	16 in octet 3	4,094

PREFIX	MASK	BLOCK SIZE	USABLE HOSTS
/21	255.255.248.0	8 in octet 3	2,046
/22	255.255.252.0	4 in octet 3	1,022
/23	255.255.254.0	2 in octet 3	510
/24	255.255.255.0	1 in octet 3	254
/25	255.255.255.128	128	126
/26	255.255.255.192	64	62
/27	255.255.255.224	32	30
/28	255.255.255.240	16	14
/29	255.255.255.248	8	6
/30	255.255.255.252	4	2 (point-to-point links)
/31	255.255.255.254	2	2, no broadcast (point-to-point only)
/32	255.255.255.255	1	A single host

- Mask octet values: 128, 192, 224, 240, 248, 252, 254, 255 = 1 to 8 network bits
- Binary place values: 128, 64, 32, 16, 8, 4, 2, 1
- Powers of two: 2, 4, 8, 16, 32, 64, 128, 256, 512, 1024, 2048, 4096
- Usable hosts = 2 to the host bits, minus 2. Subnets created = 2 to the borrowed bits
- Hosts needed: add 2, round up to the next power of two, that many host bits
- A subnet must start on a multiple of its own block size (192.168.10.96/26 is invalid; 64 and 128 are valid starts)
- Summarization: 10.1.0.0/24 through 10.1.3.0/24 = 10.1.0.0/22

Block size method

- 1. Find the interesting octet: the first mask octet that is not 255 or 0.
- 2. Block size = 256 minus that mask octet (.224 gives 32, .240 gives 16, .192 gives 64).
- 3. Subnets start at multiples of the block size: 0, 32, 64, 96, 128, 160, 192, 224 for a block of 32.
- 4. Network = the multiple at or below the host. Broadcast = one less than the next multiple. Usable = everything between.
- Example: 192.168.1.100/27. Block 32. Network 192.168.1.96, broadcast .127, hosts .97 to .126.

05 IPv6

PREFIX	TYPE	NOTES
2000::/3	Global unicast	Routable on the internet (addresses starting 2 or 3)
fe80::/10	Link-local	Automatic on every interface, never routed. Used by NDP and routing protocols
fc00::/7 (in practice fd..)	Unique local	Private addressing, routable inside the organization only
ff00::/8	Multicast	ff02::1 all nodes, ff02::2 all routers. No broadcast in IPv6
::1	Loopback	This host
::	Unspecified	No address yet
2001:db8::/32	Documentation	Used in examples only

- 128 bits, eight groups of four hex digits. A standard LAN is a /64: 64 bits network, 64 bits interface ID
- Shortening: drop leading zeros in each group; replace one run of all-zero groups with :: (only once)
- SLAAC: the router advertisement supplies the /64 prefix and the host builds its own interface ID
- EUI-64: split the 48-bit MAC, insert ff:fe in the middle, flip the seventh bit

- DHCPv6: stateful hands out addresses; stateless hands out only options such as DNS
- NDP replaces ARP: neighbor solicitation and neighbor advertisement over ICMPv6 multicast
- Dual stack runs both. Tunneling (6to4, ISATAP, Teredo, GRE) carries v6 inside v4. NAT64 with DNS64 translates for v6-only clients

06 Cabling, connectors, and Ethernet standards

COPPER	SPEED	DISTANCE	NOTES
Cat 5e	1 Gbps	100 m	Minimum for gigabit
Cat 6	1 Gbps; 10 Gbps	100 m; 55 m at 10 Gbps	
Cat 6A	10 Gbps	100 m	The answer for 10 Gbps at full distance
Cat 7	10 Gbps	100 m	Shielded, non-RJ45 connectors (GG45, TERA), rare
Cat 8	25 or 40 Gbps	30 m	Short data center runs
Coax RG-6	Cable modem, video		F-type connector; BNC on legacy LANs
Twinax (DAC)	10 to 100 Gbps	A few meters	Transceivers permanently attached; cheaper than optics

- All twisted pair Ethernet: 100 meters (328 feet) including patch cords
- RJ45 is 8P8C for Ethernet; RJ11 for telephone
- UTP default; STP for high-EMI areas (motors, fluorescent ballasts)
- T568B pins: 1 white-orange, 2 orange, 3 white-green, 4 blue, 5 white-blue, 6 green, 7 white-brown, 8 brown. T568A swaps the orange and green pairs
- Same standard both ends = straight-through (host to switch). Different = crossover (switch to switch without auto-MDIX)
- Plenum-rated for air-handling spaces above drop ceilings; riser-rated for vertical runs between floors
- CSMA/CD: listen, send, back off on collision. Full duplex has no collisions. Duplex mismatch = late collisions and CRC errors

FIBER	CORE	SOURCE	REACH	JACKET
Single-mode (SMF)	about 9 microns	Laser	10 km and beyond	Yellow
Multimode (MMF)	50 or 62.5 microns	LED or VCSEL	About 300 to 550 m at 10 Gbps	Orange (OM1, OM2), aqua (OM3, OM4)

- LC: small square snap-in, most common on transceivers. SC: larger square push-pull. ST: round bayonet twist, legacy. MPO/MTP: 12 or 24 fibers in one plug for 40 and 100 Gbps
- UPC blue flat polish; APC green angled polish (less reflection, long haul and PON). Never mate UPC to APC
- Media converter changes medium without changing data (fiber to copper, SMF to MMF)
- Transceivers: SFP 1 Gbps, SFP+ 10 Gbps, QSFP/QSFP+ 40 Gbps, QSFP28 100 Gbps, QSFP-DD and OSFP 400 Gbps. Both ends must match speed, wavelength, and fiber type

STANDARD	SPEED	MEDIUM	DISTANCE
10BASE-T	10 Mbps	Cat 3 or better	100 m
100BASE-TX	100 Mbps	Cat 5 or better	100 m
1000BASE-T	1 Gbps	Cat 5e or better	100 m
10GBASE-T	10 Gbps	Cat 6A (Cat 6 to 55 m)	100 m
40GBASE-T	40 Gbps	Cat 8	30 m
1000BASE-SX	1 Gbps	Multimode	About 550 m
1000BASE-LX	1 Gbps	Single-mode (or MMF with mode conditioning)	10 km
10GBASE-SR	10 Gbps	Multimode	About 300 m (400 m on OM4)
10GBASE-LR	10 Gbps	Single-mode	10 km

STANDARD	SPEED	MEDIUM	DISTANCE
10GBASE-ER	10 Gbps	Single-mode	40 km

Naming: S = short reach, multimode. L = long reach, single-mode. E = extended reach. T = twisted pair. SR on single-mode fiber, or SR at 2 km, is a mismatch.

07 Wireless

STANDARD	NAME	BAND	MAX RATE	KEY FEATURE
802.11a		5 GHz	54 Mbps	OFDM, short range
802.11b		2.4 GHz	11 Mbps	Long range, slow
802.11g		2.4 GHz	54 Mbps	Backward compatible with b
802.11n	Wi-Fi 4	2.4 and 5 GHz	600 Mbps	MIMO, 40 MHz channels
802.11ac	Wi-Fi 5	5 GHz only	Gigabit class (up to 6.9 Gbps)	MU-MIMO, 80 and 160 MHz channels
802.11ax	Wi-Fi 6 / 6E	2.4, 5, and 6 GHz (6E)	About 9.6 Gbps	OFDMA for crowded spaces
802.11be	Wi-Fi 7	2.4, 5, and 6 GHz	Tens of Gbps	320 MHz channels, multi-link operation

- 2.4 GHz: non-overlapping channels 1, 6, 11 (20 MHz). Longest range, most interference (microwaves, Bluetooth, neighbors)
- 5 GHz: many channels, 40/80/160 MHz bonding, DFS channels must yield to radar. 6 GHz: Wi-Fi 6E only, shortest range
- Wider channels = more speed, more overlap and noise. Use 20 MHz in dense 2.4 GHz deployments
- SSID = network name. BSSID = MAC of one radio on one AP. ESSID = same SSID across many APs for roaming
- Infrastructure (through an AP), ad hoc (IBSS, no AP), mesh (APs link wirelessly), point-to-point (directional antennas between buildings)
- Omnidirectional for general coverage; directional (Yagi, patch, parabolic) for long links and hallways
- Autonomous APs configured individually; lightweight APs managed by a wireless LAN controller
- Site survey produces a heat map. Overlap cells 15 to 20 percent for roaming. Band steering pushes clients to 5 GHz

SECURITY	STATUS	DETAIL
WEP	Broken	Never use
WPA	Deprecated	TKIP
WPA2	Acceptable	AES with CCMP. Personal (PSK) or Enterprise (802.1X with RADIUS)
WPA3	Current	Personal uses SAE (no offline cracking). Enterprise offers 192-bit mode
Enterprise		802.1X authentication against RADIUS, per-user credentials or certificates, per-user VLAN
Captive portal		Intercepts first web request for terms or login; guest SSID on an isolated VLAN
Hidden SSID, MAC filtering	Not security	Trivially bypassed; hardening at best

08 Switching

- Switch jobs: learn source MACs into the MAC (CAM) table, forward known destinations, flood unknowns and broadcasts, block loops with STP
- Each switch port is one collision domain. Each VLAN is one broadcast domain. A hub is one collision domain. A router separates broadcast domains
- Unmanaged: no config. Managed: VLANs, STP, port security. Layer 3 switch: also routes between VLANs in hardware

- In-band management over the network (SSH, web). Out-of-band via console port or dedicated management network
- Access port: one VLAN, untagged. Trunk port: many VLANs tagged with 802.1Q (4-byte tag, IDs 1 to 4094)
- Native VLAN crosses the trunk untagged (default VLAN 1) and must match on both ends
- Voice VLAN lets a phone tag while the PC behind it stays untagged. Management VLAN holds switch interfaces
- Routing between VLANs: SVI on a Layer 3 switch, or router on a stick with one subinterface per VLAN
- LACP (802.3ad, 802.1AX) bundles links; every member needs matching speed, duplex, and VLANs. One flow still uses one link
- MTU 1500; jumbo about 9000; every device in the path must match. Symptom of mismatch: small pings work, large transfers fail
- Port mirroring (SPAN) copies traffic to an analyzer port. Port security limits MACs; violation puts the port in error-disabled
- Duplex mismatch: one side hard-coded, the other auto falls to half duplex. Late collisions, CRC errors, slow link

SPANNING TREE	DETAIL
Standard	STP 802.1D; RSTP 802.1w; MSTP 802.1s
Root bridge	Lowest bridge ID (priority then MAC). Set core priority low
Port roles	Root port (toward root), designated port (one per link), blocked (redundant). RSTP adds alternate and backup
STP states	Blocking, listening, learning, forwarding: 30 to 50 seconds
RSTP states	Discarding, learning, forwarding: converges in seconds
PortFast	Skips listening and learning on host ports. Never on switch-to-switch links
BPDU guard	Shuts a PortFast port that receives a BPDU (rogue switch)
Root guard	Stops a downstream switch from becoming root
Loop guard	Protects against unidirectional link failures
Loop symptoms	Broadcast storm, CPU at 100 percent, MAC addresses flapping, everything slow after adding a redundant cable

POE STANDARD	NAME	AT THE PORT	DELIVERED TO DEVICE
802.3af	PoE	15.4 W	About 13 W
802.3at	PoE+	30 W	About 25.5 W
802.3bt Type 3	PoE++	60 W	About 51 W
802.3bt Type 4	PoE++	90 W	About 71 W

09 Routing, NAT, and gateway redundancy

Route selection order

- 1. Longest prefix match wins (a /24 beats a /16 beats 0.0.0.0/0).
- 2. Same prefix: lowest administrative distance wins.
- 3. Same source: lowest metric wins.

ROUTE SOURCE	ADMINISTRATIVE DISTANCE
Directly connected	0
Static	1
External BGP	20
EIGRP	90
OSPF	110
IS-IS	115

ROUTE SOURCE		ADMINISTRATIVE DISTANCE	
RIP		120	
Internal BGP		200	

PROTOCOL	TYPE	METRIC	NOTES
RIP	Distance-vector, IGP	Hop count, max 15	Updates every 30 s; RIPv2 supports CIDR; RIPng for IPv6; small networks
OSPF	Link-state, IGP	Cost from bandwidth	Open standard, areas, Dijkstra SPF, fast convergence, multicast 224.0.0.5 and .6
EIGRP	Advanced distance-vector (hybrid), IGP	Bandwidth and delay	Cisco origin, fast convergence
BGP	Path-vector, EGP	AS path and policy	Between autonomous systems, runs on TCP 179, the internet's protocol, slow by design

- Distance-vector knows what neighbors say; link-state knows the whole map (more CPU, faster, fewer loops)
- Static: predictable, no overhead, no adaptation. Dynamic: adapts to failures. Default route 0.0.0.0/0 is the way out
- Routers rewrap the frame (new MACs), keep the packet (same IPs), decrement TTL by one. TTL 0 = discard
- Static NAT: one to one, publish a server. Dynamic NAT: pool, many to many. PAT (overload): many to one by source port, every home router
- Port forwarding: public port to an inside host. Inside local = private address; inside global = its public translation
- FHRP: routers share a virtual IP and virtual MAC. HSRP Cisco (active/standby). VRRP open standard (master/backup). GLBP Cisco with load balancing
- Subinterfaces: one physical interface split per VLAN (router on a stick)
- Convergence: time for all routers to agree after a change

10 DHCP, DNS, and time

DHCP

- DORA: Discover (broadcast from 0.0.0.0 to 255.255.255.255), Offer, Request, Acknowledge. UDP 67 server, 68 client
- Scope = the pool. Exclusion = addresses never leased. Reservation = fixed address pinned to a MAC. Lease = how long
- Clients renew at 50 percent of the lease (T1) and rebind at 87.5 percent (T2)
- Options: 3 default gateway, 6 DNS servers, 15 domain name, 42 NTP, 66 TFTP server name, 150 Cisco phone TFTP
- Relay agent or IP helper address turns the broadcast into a unicast so one server can serve many subnets
- Scope exhaustion or unreachable server = client self-assigns 169.254.x.x (APIPA). Rogue DHCP = DHCP snooping

DNS RECORD	MAPS
A	Name to IPv4 address
AAAA	Name to IPv6 address
CNAME	Alias: one name to another name
MX	Mail exchanger for the domain, with priority
NS	Authoritative name servers for a zone
SOA	Start of authority: zone serial and timers
PTR	IP address back to a name (reverse zone, in-addr.arpa)
TXT	Free text: SPF, DKIM, DMARC, domain verification
SRV	Locates a service by name, port, and priority (SIP, Active Directory)

- Lookup path: recursive resolver, root, top-level domain, authoritative. Cached answers are non-authoritative and expire at the TTL
- Forward zone: names to addresses. Reverse zone: addresses to names. Primary holds the writable copy; secondaries pull by zone transfer on TCP 53
- DNSSEC signs records: integrity, not privacy. DoT (TCP 853) and DoH (443) encrypt queries: privacy
- The hosts file is checked before DNS
- NTP UDP 123, millisecond accuracy. Stratum 0 = reference clock (GPS, atomic); stratum 1 connects to it; stratum 2 syncs from 1
- PTP (IEEE 1588) = sub-microsecond with hardware timestamps. NTS = authenticated, encrypted NTP
- Clock skew over 5 minutes breaks Kerberos; wrong time breaks certificates, MFA codes, and log correlation

11 Appliances, cloud, and modern networking

DEVICE OR FUNCTION	LAYER	ONE-LINE JOB
Router	3	Connects networks, chooses paths, separates broadcast domains
Switch	2	Forwards frames by MAC; one collision domain per port
Layer 3 switch	2 and 3	Switch that routes between VLANs in hardware
Access point, WLC	2	Bridges Wi-Fi to the LAN; the controller manages many APs
Firewall	3 to 7	Stateful tracks connections; next-generation inspects applications and users
IDS		Watches a copy of traffic, alerts only. Passive, out of band
IPS		Inline in the path, blocks in real time. Can bottleneck or false-positive
VPN concentrator		Terminates many encrypted tunnels
Forward proxy	7	Requests on behalf of clients; filters, caches, hides client addresses
Reverse proxy	7	Sits in front of servers; terminates TLS, caches, protects
Load balancer	4 to 7	Spreads connections across servers: round robin, least connections, health checks
CDN		Caches content near users; cuts latency, absorbs spikes
NAS		File-level storage over SMB or NFS
SAN		Block-level storage over iSCSI or Fibre Channel; looks like a local disk
QoS		Classifies and prioritizes traffic so voice and video survive congestion

Cloud

- SaaS: use the application. PaaS: deploy your code, provider runs the OS and runtime. IaaS: rent VMs, you manage the OS
- Public (shared, pay as you go), private (dedicated), hybrid (mix)
- VPC: your isolated virtual network with its own subnets
- Network security group: stateful firewall per VM interface. Network security list or NACL: stateless rules per subnet
- Internet gateway: public addresses reach the internet. NAT gateway: private subnets reach out only
- NFV: routers, firewalls, load balancers as software instances
- Site-to-site VPN over the internet (cheap, variable) versus direct connect (private circuit, predictable, costly)
- Scalability grows capacity; elasticity grows and shrinks automatically; multitenancy shares hardware with logical isolation

SDN, SD-WAN, VXLAN, zero trust, IaC

- SDN separates the control plane (decisions) from the data plane (forwarding). Application plane on top, management plane for monitoring
- Northbound APIs: applications to controller. Southbound APIs: controller to devices
- SD-WAN: application aware, transport agnostic, zero-touch provisioning, central policy management
- VXLAN: Layer 2 frames inside UDP across Layer 3. 24-bit VNI = 16 million segments versus 4094 VLANs. Data center interconnect
- Zero trust: never trust, always verify. Every request authenticated and authorized by policy with least privilege

- SASE = SD-WAN plus cloud security (firewall, secure web gateway, ZTNA, CASB). SSE = the security half only
- IaC: playbooks and templates, source control, drift detection, dynamic inventories, automated upgrades

Physical installation

- MDF: main wiring room, core devices, demarcation point where the provider's responsibility ends
- IDF: closet per floor or wing with access switches, uplinked to the MDF
- 19-inch racks; 1U = 1.75 inches. Two-post for light gear, four-post or lockable cabinet for servers
- Front intake cold aisle, rear exhaust hot aisle. Match port-side intake or exhaust to the rack's airflow
- Patch panel terminates horizontal cabling; patch cords connect panel to switch. Label both ends
- UPS carries load through outages and conditions power; generator is long term. PDU is the rack power strip
- Temperature 18 to 27 C (64 to 80 F). Humidity 40 to 60 percent: too dry = static, too humid = condensation
- Clean-agent or gas fire suppression, not water. 120 V or 208/240 V; redundant supplies on separate circuits

12 Network operations

Documentation and agreements

- Physical diagram: where devices sit and cables run. Logical diagram: subnets, VLANs, routing, traffic flow
- Layer 1 diagram cabling; Layer 2 switches, VLANs, trunks, STP; Layer 3 routers, subnets, routing protocols
- Rack diagram: front view by rack unit. Cable map: jack to patch panel port to switch port
- Asset inventory: model, serial, location, owner, warranty, licensing. IPAM: every subnet and address
- SLA: what the provider guarantees (uptime, repair time) and penalties. MOU: non-binding intent. NDA: confidentiality
- Update documentation as the last step of every change

Life cycle, change, and configuration

- EOL: vendor stops selling. EOS: no more patches or support; replace it
- Change process in order: request, assess impact and write a rollback plan, approve, schedule a maintenance window and notify, implement and test, document
- Emergency changes take a shorter path but are still recorded. Service request = routine pre-approved work
- Production config (running now), backup config (saved before and after changes), baseline or golden config (approved standard). Drift = deviation from golden

MONITORING	DETAIL
SNMP	Agent on the device, MIB database, OID per variable. Manager polls with Get on UDP 161; device sends traps on UDP 162
SNMP versions	v1 and v2c: plaintext community string (v2c adds bulk). v3: authentication, integrity, encryption
Flow data	NetFlow, sFlow, IPFIX: who talked to whom, ports, volume. No payload. Finds top talkers
Packet capture	Protocol analyzer (Wireshark) fed by a SPAN port or an inline tap. Every byte
Syslog severity	0 Emergency, 1 Alert, 2 Critical, 3 Error, 4 Warning, 5 Notice, 6 Informational, 7 Debug
SIEM	Correlates logs from many sources, alerts on patterns, retains for compliance
Baseline	Normal utilization, latency, errors over time. Anomaly alerts fire on deviation
Solutions	Discovery (ad hoc or scheduled), traffic analysis, performance, availability (uptime versus SLA), configuration monitoring

Metrics

- Bandwidth = capacity. Throughput = what you get. Utilization = the ratio; sustained above 70 to 80 percent means congestion
- Latency = delay in ms (distance, queueing, satellite adds 500 to 600 ms). Jitter = variation in delay; choppy voice
- Packet loss = missing packets: TCP retransmits, voice gaps. Interface errors: CRC, runs, giants, discards
- Device health: CPU, memory, temperature, fans. CPU pegged often means a broadcast storm or attack

DISASTER RECOVERY	DETAIL
RPO	Recovery point objective: how much data you can lose, measured backward in time
RTO	Recovery time objective: how fast service must be restored
MTTR	Mean time to repair: average fix time
MTBF	Mean time between failures: average life before failing
Cold site	Space, power, cooling only. Days or weeks. Cheapest
Warm site	Equipment and connectivity in place; restore data. Hours to a day
Hot site	Running mirror with current data. Minutes. Most expensive
Active-active	All nodes serve; capacity drops on failure
Active-passive	Standby takes over on failure (FHRPs)
Tabletop	Walk through the plan on paper. Validation test: actually fail over or restore

Remote access and management

- Site-to-site VPN joins offices. Client-to-site joins remote users. Clientless = browser over TLS
- Split tunnel: only corporate traffic through the VPN. Full tunnel: everything, inspected and protected
- Manage by SSH (22), HTTPS GUI, API, or console cable. Jump box or bastion: one hardened host you reach everything through
- In-band uses the production network. Out-of-band (console server, management network, cellular) works when production is down

13 Security

Concepts

- CIA: confidentiality (encryption, access control), integrity (hashing, signatures), availability (redundancy, DDoS protection)
- Vulnerability = weakness. Threat = what could exploit it. Exploit = the method. Risk = likelihood times impact
- Encryption in transit (TLS, IPsec, SSH) versus at rest (disk, database). PKI: a certificate authority binds a public key to an identity. Self-signed = browser warning
- AAA: authentication (who), authorization (what), accounting (record)
- MFA factors: something you know, have, are; plus location and time. Password plus phone app code = know plus have
- SSO: one login for many systems. SAML: web SSO between identity provider and applications
- RADIUS: UDP 1812/1813, network access (Wi-Fi, VPN, 802.1X), encrypts only the password
- TACACS+: TCP 49, device administration, encrypts the whole payload, separates authentication from authorization, per-command control
- Least privilege; role-based access control; geofencing
- Honeypot = decoy system; honeynet = decoy network. Physical: locks, badges, cameras, mantraps
- Data locality laws; PCI DSS for card data; GDPR for EU personal data. Segment IoT, IIoT, SCADA/ICS/OT, guest, and BYOD
- IPsec: AH = integrity and authentication only; ESP = encryption too. Transport mode encrypts the payload; tunnel mode wraps the packet (site-to-site). IKE on UDP 500

SYMPTOM	ATTACK	DEFENSE
Service flooded from one source, or from many (botnet)	DoS, DDoS (reflection and amplification)	Upstream filtering, DDoS protection
Attacker reaches a VLAN they should not; switch spoofing or double 802.1Q tags	VLAN hopping	Disable trunk negotiation on access ports, unused native VLAN
Switch forwards all traffic to all ports; MAC table full of fakes	MAC flooding	Port security

SYMPTOM	ATTACK	DEFENSE
Forged ARP replies map the gateway IP to the attacker's MAC; traffic flows through them	ARP poisoning enabling an on-path attack	Dynamic ARP inspection
Users sent to attacker sites by forged answers or corrupted cache	DNS poisoning or spoofing	DNSSEC
Clients get a wrong gateway or DNS from an unknown server	Rogue DHCP	DHCP snooping
Forged source address	IP spoofing	Ingress filtering
Unauthorized AP on the network	Rogue access point	Wireless IDS, NAC, port security
Attacker AP broadcasting your SSID	Evil twin	Enterprise authentication, user awareness
Clients kicked off with forged management frames	Deauthentication	802.11w protected management frames
Email, phone, text, targeted, following through a door, watching a screen, trash	Phishing, vishing, smishing, spear phishing, tailgating, shoulder surfing, dumpster diving	Training, mantraps, shredding
Encrypts data for payment; hides in software; self-propagates; watches	Ransomware, trojan, worm, spyware	Backups, patching, endpoint protection
Every password tried; common words; leaked credentials reused	Brute force, dictionary, credential stuffing	MFA, lockout, unique passwords

Hardening and defense

- First steps on any device: change default passwords, disable unused ports and services, patch firmware, SSH and HTTPS only, management on its own VLAN or out of band
- Rotate keys and certificates; revoke compromised ones
- NAC: 802.1X authentication plus posture assessment (antivirus, patches, disk encryption). Noncompliant devices go to a quarantine or remediation VLAN
- ACLs: processed top down, first match wins, implicit deny at the end
- URL filtering by address or category; content filtering inspects what is inside
- Firewall zones: trusted (inside), untrusted (internet), screened subnet (DMZ) for public-facing servers
- Segmentation with VLANs and firewall rules limits lateral movement

14 Troubleshooting

The seven steps, in order

- 1. Identify the problem: gather information, question users, identify symptoms, determine what changed, duplicate it, approach multiple problems individually.
- 2. Establish a theory of probable cause: question the obvious; top-to-bottom, bottom-to-top, or divide and conquer through the OSI layers.
- 3. Test the theory to determine the cause. Not confirmed: new theory or escalate.
- 4. Establish a plan of action to resolve the problem and identify potential effects.
- 5. Implement the solution or escalate as necessary.
- 6. Verify full system functionality and, if applicable, implement preventive measures.
- 7. Document findings, actions, outcomes, and lessons learned.

After the theory is confirmed, the next step is the plan, not the fix. After implementing, verify before documenting. Documenting is always last.

SYMPTOM	CAUSE
CRC errors and late collisions on one link	Duplex mismatch
Works at 80 m, fails or flaps at 120 m	Attenuation (copper limit 100 m)
Fiber link dead, strands crossed at one end	TX/RX polarity reversal
Runts (under 64 bytes)	Collisions or bad NIC
Giants (over MTU)	MTU or jumbo frame mismatch
Discards or drops rising	Congestion
Port error-disabled	Port security, BPDU guard, or flapping violation; fix cause then clear
Port administratively down	Someone typed shutdown
Camera or AP powers on then reboots, or never powers	PoE budget exceeded or wrong PoE standard
No link with SR optic on single-mode fiber	Transceiver or fiber mismatch
Interference in a factory over UTP	EMI; use STP or fiber
Broadcast storm, CPU 100 percent, MACs flapping right after a new cable	Spanning tree loop
Odd traffic paths, access switch is root	Root bridge in the wrong place; lower core priority
Host gets an address from the wrong scope after a desk move	Access port in the wrong VLAN
Untagged traffic lands in the wrong VLAN on a trunk	Native VLAN mismatch
Local devices reachable, nothing remote	Wrong or missing default gateway
Some hosts on the same subnet unreachable	Incorrect subnet mask
Intermittent connectivity, address conflict warning	Duplicate IP (static inside a scope without exclusion)
New clients get 169.254.x.x, old ones work	DHCP scope exhaustion or unreachable server
Ping by IP works, by name fails	DNS
One application or port fails, others work	ACL blocking it
Traffic goes out, replies never come back	Missing return route or asymmetric routing through a stateful firewall
Full signal bars, poor throughput	Interference or co-channel overlap; use 1, 6, 11 or move to 5 GHz
Choppy voice, delay varies	Jitter
Everything slow through one link at 100 percent	Bottleneck or congestion
Clients cling to a distant AP	Too little cell overlap or transmit power too high

COMMAND OR TOOL	SHOWS
ping	Reachability, round-trip time, loss. Loopback, self, gateway, remote in that order
tracert / traceroute	Every hop and its latency; where the path breaks. mtr and pathping repeat over time
ipconfig /all, /release, /renew, /flushdns	Windows address, mask, gateway, DNS, lease, MAC; refresh lease; clear resolver cache
ifconfig, ip addr	Linux and macOS addressing
arp -a	IP-to-MAC cache; duplicates and poisoning
nslookup, dig	Query DNS directly, pick a server, ask for a record type
netstat	Open connections and listening ports; netstat -r shows the routing table
nmap	Open ports and services; live hosts on a subnet (authorized use only)
tcpdump, Wireshark	Command-line capture; graphical protocol analyzer

COMMAND OR TOOL	SHOWS
LLDP / CDP	What device and port is on the other end of a cable
show mac-address-table	Which MAC is on which port and VLAN
show arp	The device's ARP cache
show interface	Status, speed, duplex, errors, CRCs, drops
show ip route	Routing table and how each route was learned
show vlan	VLANs and port assignments
show running-config	Current configuration
show power	PoE budget and per-port power
HARDWARE TOOL	USE
Toner and probe	Find the far end of a cable in a bundle or patch panel
Cable tester	Continuity, pinout, opens, shorts, split pairs, length
Cable certifier	Proves a run meets its category with a report
Punch-down tool, crimper	Terminate on jacks and panels; attach RJ45 plugs
Loopback plug	Tests a port's own transmit and receive
Visual fault locator	Red light shows fiber breaks and bends
OTDR	Locates fiber breaks and losses by distance
Optical power meter	Received light level
Fiber cleaning kit	Dirty connectors are the most common fiber fault
Network tap	Passive inline copy of traffic when there is no SPAN port
Wi-Fi analyzer	APs, channels, signal, dead spots
Spectrum analyzer	All radio energy including non-Wi-Fi interference
Multimeter	Voltage and continuity
PoE injector	Adds power where the switch cannot